

Table 1
Summary of boiling metrics for different surfaces tested

	Surface	CHF (W/cm ²) (% change)	Incipience superheat (°C) (% change)	Fully developed boiling superheat (°C) (% change)	$h_{\max}$ (W/m ² K) (% change)
<i>Silicon</i>					
Smooth silicon	Sample A	12.5	23.7	31.9	3,299
	Sample B	11.8	39.5	28.5	3,197
	Sample C	11.4	23.3	33.2	3,104
Fully CNT-coated silicon	Light array, test 1	18.1 (44.8%)	7.9 (−66.7%)	10.2 (−68.0%)	18,200 (452%)
	Light array, test 2	17.1 (36.8%)	7.1 (−70.0%)	9.3 (−77.9%)	15,732 (377%)
	Dense array, test 1	13.3 (6.4%)	7.2 (−69.6%)	8.4 (−73.7%)	12,376 (275%)
	Dense array, test 2	13.3 (6.4%)	7.9 (−66.7%)	8.3 (−74.0%)	13,348 (305%)
Patterned CNT on silicon	‘Grid’ array, test 2	15.8 (26.4%)	12.7 (−46.4%)	12.7 (−60.2%)	12,700 (285%)
	‘Grid’ array, test 1	15.0 (20.0%)	21.2 (−10.6%)	21.2 (−33.5%)	11,800 (258%)
	‘Island’ array, test 2	14.4 (15.2%)	15.6 (−34.2%)	18.0 (−43.6%)	4,562 (28.3%)
	‘Island’ array, test 1	13.6 (8.8%)	14.0 (−40.9%)	16.1 (−49.5%)	6,097 (84.8%)
<i>Copper</i>					
Bare copper	Test 2	18.4	13.9	13.9	13,200
	Test 1	17.7	7.2	7.2	9,540
Fully CNT-coated copper	Test 2	19.5 (6.0%)	8.4 (−39.6%)	13.9 (0%)	17,510 (32.7%)
	Test 1	19.4 (5.4%)	7.9 (−43.2%)	8.8 (−36.7%)	17,111 (29.6%)
Uncoated copper microstud	Test 2	32.3 (75.5%)	9.4 (−32.4%)	11.5 (−17.3%)	10,800 (−18.2%)
	Test 1	31.9 (73.4%)	9.1 (−34.5%)	10.6 (−23.7%)	11,900 (−9.9%)
CNT-coated copper microstud	Test 1	26.0 (41.3%)	5.5 (−60.4%)	9.0 (−35.3%)	26,500 (100.8%)
	Test 2	25.2 (37.0%)	5.1 (−63.3%)	9.0 (−35.3%)	23,930 (81.3%)

Comparisons for silicon surfaces are referenced to smooth surface sample A. Comparisons for copper surfaces are referenced to bare copper test 2. Reference conditions for each material are those that yielded the highest CHF value in repeatability tests.

expense of a noticeable (20%) reduction in CHF. Boiling incipience and fully developed boiling superheats were improved by 42% and 22%, respectively, and the maximum heat transfer coefficient by 145% over the uncoated microstud surface. The relatively poor CHF performance of the CNT-coated surface may be partially explained by the absence of CNTs along the circumference of the microstuds. Additionally, the thermal resistance of the copper microstuds themselves appears to reduce the effectiveness of the top CNT-coated surfaces of the microstuds compared to the CNT-coated base area between microstuds. Near CHF, the heat flow is biased to the CNT-coated base area, inducing severe vapor coalescence in the base region that engulfs the circumference and top surfaces of the microstuds.

Two separate tests with the uncoated microstud surface showed 8% deviation in fully developed boiling superheat and 1% difference in CHF. Repeatability tests with the CNT-coated microstud surface showed zero deviation in fully developed boiling superheat and 3% in CHF.

3.4. Summary of performance metrics

Table 1 provides a summary of boiling performance metrics for the different surface materials, CNT array densities and CNT area coverages tested in the present study. Performance parameters for different silicon tests are compared in absolute and percent change terms to those for

bare silicon. Similarly, copper tests are compared to bare copper. A clear trend emerges from this table. Excepting the fully-coated ‘dense’ CNT array case (which provided slightly lower superheat but lower CHF values than the fully-coated ‘light’ CNT array case), best *overall* results (lowest incipience superheat, lowest fully developed boiling superheat, highest boiling heat transfer coefficient, and highest CHF) were achieved by fully coating surfaces with a ‘light’ CNT array. Furthermore, the relative improvements are far more impressive for silicon because, as mentioned earlier, the reference uncoated silicon surface was very smooth and free of sizeable nucleation sites to start with.

4. Conclusions

This work has examined the effectiveness of carbon nanotubes (CNTs) in enhancing the pool boiling performance of low-contact-angle dielectric liquid FC-72 on silicon and copper surfaces similar to those used in the electronics industry. Such liquids pose several challenges to the implementation of pool boiling schemes, including large and erratic incipient boiling superheat and low CHF. Different CNT array densities and area coverages were tested to address these shortcomings. Boiling curves were generated to determine key performance metrics for different substrate materials and CNT patterns. Key findings from this study follow.